JioMart Heuristic Evaluation

UX Case Study

Central UI/UX Research Team

About JioMart App

JioMart, also known as DESH KI NAYI DUKAAN, is Reliance Retail's new e-commerce initiative that was launched in 2020 with an aim to integrate digital and physical retail stores to help Indian consumers with unlimited choices, superior value proposition, and unmatched shopping experience by digitally enabling and empowering merchant partners.

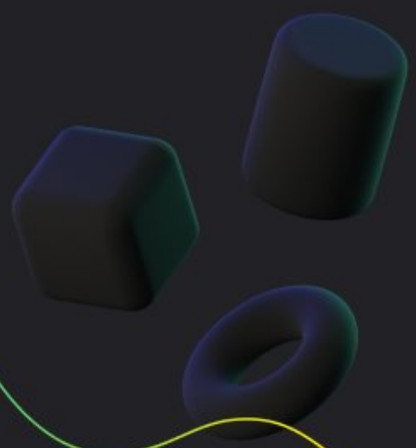
Overview

In design, Heuristic Analysis is a process where experts use rules of thumb to measure the usability of user interfaces in independent walkthroughs and report issues or in other words, it is a way to test whether a website is user friendly.

Heuristic ⟶ means "enabling a person to discover or learn something for themselves".

JioMart App

10 Usability Heuristics for User Interface Design



Central UI/UX Research Team

01

Visibility of system status

Always keep users informed about what is going on, through appropriate feedback within reasonable time.

02

Match between system and the real world

Follow real-world conventions, making information appear in a natural and logical order.

03

User control and freedom

Users should leave the unwanted state without having to go through an extended dialogue. undo and redo.

04

Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing.

05

Error prevention

Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action

06

Recognition rather than recall

Minimize the user's memory load by making objects, actions, and options visible.

07

Flexibility and efficiency of use

Accelerators. Allow users to tailor frequent actions.

08

Aesthetic and minimalist design

Dialogues should not contain information which is irrelevant or rarely needed.

09

Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

10

Help and documentation

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation.

Evaluation of Heuristics

To further examine the impact of each of the problems, we estimated both its severity in terms of usability principles and the ease with which the problem might be solved.

Severity Rating Scale

We scored each usability issue on a scale of 0 to 4, with 0 being no issue and 4 being catastrophic. We looked at three factors to determine the severity - frequency, impact, and persistence.

Rating	Definition
S0	Violates a heuristic but doesn't seem to be a usability problem.
S1	Superficial usability problem: may be easily overcome by the user or occurs extremely infrequently. Does not need to be fixed in the next iteration unless extra time is available.
S2	Minor usability problem: may occur more frequently or be more difficult to overcome. Fixing this should be given low priority for the next iteration
S3	Major usability problem: occurs frequently and persistently or users may be unable or unaware of how to fix the problem. Important to fix by the next iteration, so should be given high priority.
S4	Usability catastrophe: Seriously impairs use of product and cannot be overcome by users. Imperative to fix this by the next iteration before the product can be released.

Ease of Fixing Rankings

Rating	Definition
E0	Problem would be extremely easy to fix. Could be completed by one team member before the next iteration.
E1	Problem would be easy to fix. Involves specific interface elements and the solution is clear.
E2	Problem would require some effort to fix. Involves implementation of changes to multiple aspects of the interface before the next iteration.
E3	Usability problem would be difficult to fix. Requires concentrated development effort to finish before next iteration, involves implementation of changes to multiple aspects of the interface. Solution may not be immediately obvious and/or may be disputed.

Executive Summary

This report provides a detailed analysis of the heuristic evaluation process used to evaluate the JioMart app.

The evaluation itself was performed using the heuristic evaluation usability method, based on heuristics provided by Jakob Nielsen. This method consists of evaluators comparing a predefined set of usability principles to an application or website while attempting to complete a system task.

For this project, ten heuristics were used, focusing on the core functionalities of JioMart mobile application: browsing products, searching for deals and promotions, getting rewards points, checking of online exclusive products and sizes, checking of in-store availability.

The goal of this evaluation was to identify major usability flaws within the JioMart interface through the application.

The processes of completing the tasks were evaluated from a user perspective based on the usability heuristics from Jakob Nielsen's 10 Heuristics for User Interface Design.

The usability problems found from this evaluation were then ranked according to their severity and ease with which the problem can be solved.

Here is a summary of the key Problem Areas identified, arranged by Severity and Ease of Fixing:

Sr.No	Problem Area	Heuristic(s)	Severity Rating	Ease of Fixing Rating
1	Sign up & Sign In	#4:Consistency and standards	S0	E0
2	All category Icons	#6: Recognition rather than recall #4:Consistency and standards	S0	E0
3	List & Scan Icon	#6 :Recognition rather than recall	S0	E0
4	List feature	#2: Match between system and the real world #3: User control and freedom	S3	E2
5	Search page	#3 : User control and freedom #6 : Recognition rather than recall	S1	E1
6	Scan feature	#6 :Recognition rather than recall #5 :Error prevention #9 :Help and documentation	S3	E3
7	Voice feature	#7 : Flexibility and efficiency of use #10 : Help and documentation	S2	E2
8	Shop from top categories	#8 : Aesthetic and minimalist design #4 : Consistency and standards	S0	E0
9	Category & sub category showcase	#7 : Flexibility and efficiency of use	S2	E1
10	Top Deals	#5 : Error prevention	S2	E1
11	Address	#6 :Recognition rather than recall	S2	E1
12	Profile & Account	#4 : Consistency and standards	S2	E2
13	Mylist	#4:Consistency and standards #2:Match between system and the real world	S3	E2
14	Profile and Accoiunt	#4:Consistency and standards #2:Match between system and the real world	S4	E3
15	Scan feature	#2: Match between system and real world #4: Consistency and standards #6: Recognition rather than recall #8: Aesthetic and minimalist design	S3	E1

Sr.No	Problem Area	Heuristic(s)	Severity Rating	Ease of Fixing Rating
16	Filter & Sort	#10:Help users recognize, diagnose & recover errors	S2	E2
17	Checkout Process	#4:Consistency and standards #5:Error prevention #10:Help users recognize, diagnose recover from errors	S3	E2
18	Banner	#2:Aesthetic and minimalist design #4:Consistency and standards	S2	E2
19	Cart	#5 : Error prevention	S2	E3
20	Order details	#5:Error prevention #1:Match between system and the real world	S3	E2
21	PDP	#5: Error prevention #3: User control and freedom #4: Consistency & standard	S3	E2
22	PLP	#5 : Error prevention	S2	E2
23	End of scroll	#7:Flexibility and efficiency of use	S1	E1

Home Page : All category

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
All category Icons	#6: Recognition rather than recall #4: Consistency and standards	S0	E0

User Goal:

To search for an entity.

Elaboration of Problem:



The icons and colors of top navigational are not consistent
The icons are not intuitive and background colour looks dull

Identified Issues

No Consistency in icons and colours in all category tray

Home Page : List and scan icon

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
List & Scan Icon	#6 :Recognition rather than recall	S0	E0

User Goal:

To search for an entity.

Elaboration of Problem:



Product discovery helps the users in finding the right product. It can be through search, scan, voice commands, browse, etc

The search bar allows users to input a list of products for searching at once which matches the real-world model where users would make a shopping list.

The list & scan icon in the search is little hard to associate it with shopping list for non-tech savvy users.

Identified Issues

Placement and recognition of icons

Shop from Top Categories

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Shop from top categories	#8 : Aesthetic and minimalist design #4 : Consistency and standards	S0	E0

User Goal:

To search for an entity.

Elaboration of Problem:





Category and Sub-category section

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Category & sub category showcase	#7 : Flexibility and efficiency of use	S2	E1

User Goal:

To search for an entity.

Elaboration of Problem:



The category showcase increase the scroll length and visual representation can be relooked at.

Identified Issues

The visual representation and colour should be relooked

The category widget increase the scroll length

JioMart banner

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Banners	#2:Aesthetic and minimalist design #4:Consistency and standards	S2	E2

User Goal:

To click on banner to avail offers























- Randonm font, font size and font type used for the banner
- Absence visual brek between banner and products
- So many colour used that decrease brand
- Random placement of text and images that break visual consistency
- The main banner creatives consist of more than 7 carousals cards.

Voilate
Consistency and standards
Aesthetic and minimalist design

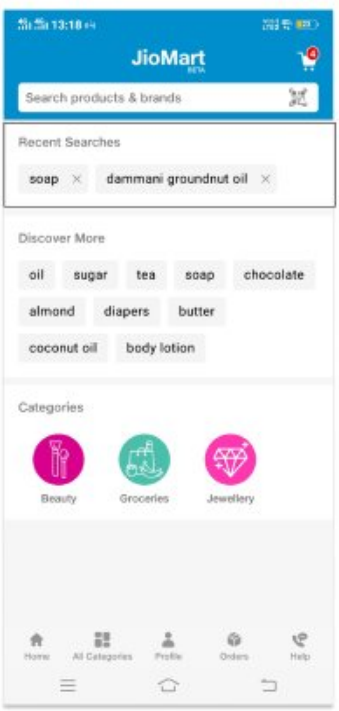
Home Page : Search page

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Search page	#3 : User control and freedom #6 : Recognition rather than recall	S1	E1

User Goal:

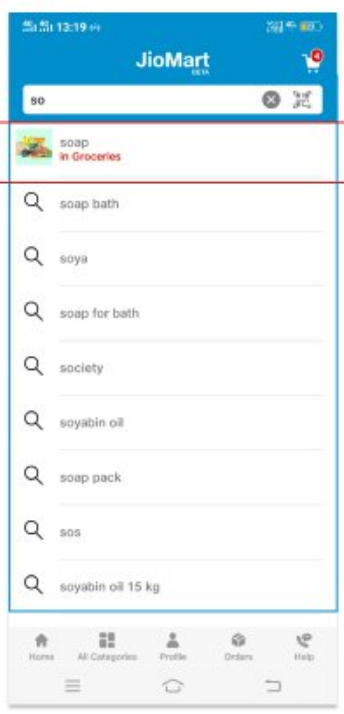
To search for an entity.

Elaboration of Problem:



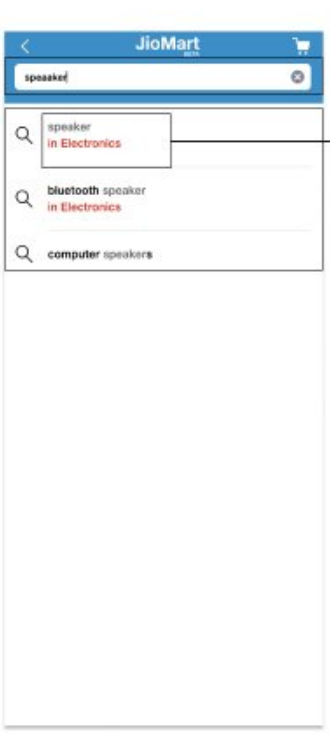
The search will take the user to a new page where he can see recent searches,discover more and categories

The user can use the search bar option to enter the product they want to find.



The image should be relooked

The category name is shown in Red colour



Used JioMart primary colour instead of red color to the main category that reflect JioMart brand

Search result with user's wrong input

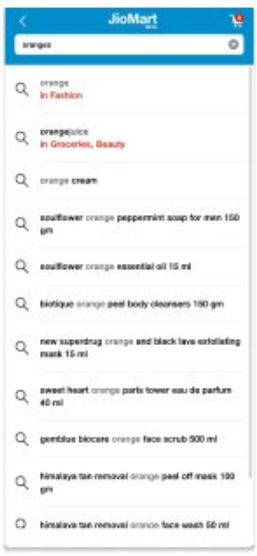


Showing all results for speaaker instead of

Showing results for speaker
Search instead for speaaker

Showing nearest possible product at wrong search keyword when user's wrong input instead of showing no result found





Identified Issues

After tapping on the search bar, the result should help user in selecting the right product

Products are shown with wrong category or sub category. For example Orange is shown in Fashion.

Most of the product showing in Groceries only

The text result is very monotonous

Recommendations

The Search algorithm can be modified to improve the list search option.

Assessing whether searching and observing too many entities belonging to different domains at once does not increase the cognitive load on the user's brain has to be checked before implementation.

QR Search Option

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Scan feature	#6 :Recognition rather than recall #5 :Error prevention #9 :Help and documentation	S3	E3

User Goal:

To search for an entity.

Elaboration of Problem:



The presence of QR Code scanner is not Clear.

The icon looks confusing to users



Presence of an image search option could be more appropriate and can be replaced/added in the application.



The results is not shown the screen gets blank

Identified Issues

The QR Code is not working.

The Photo scan is not working and the result shows a blank screen

Recommendations:

The QR Code scanner could be replaced by an image search feature after the assessment of its use, if it seems to be less beneficial for the user.

Voice Search Option

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Voice feature	#7 : Flexibility and efficiency of use #10 : Help and documentation	S2	E2

User Goal:

To search for an entity.

Elaboration of Problem:



The mic icon is in the top right corner of the screen lying beside the search bar. This increases the visibility of the icon and indicates that the mic icon can alternatively be used to search.



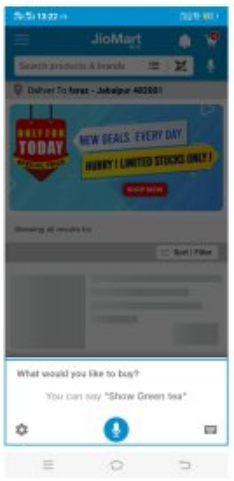
Setting panel



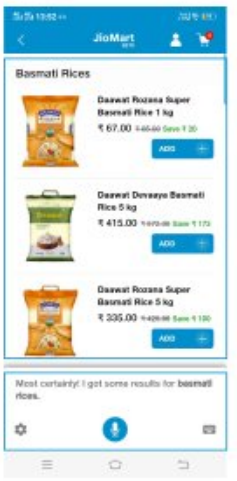
Assistance surface

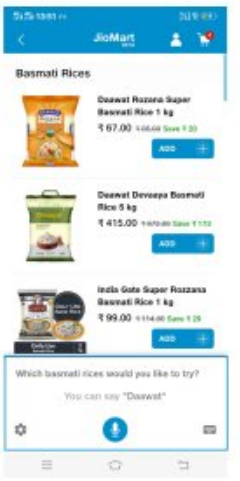


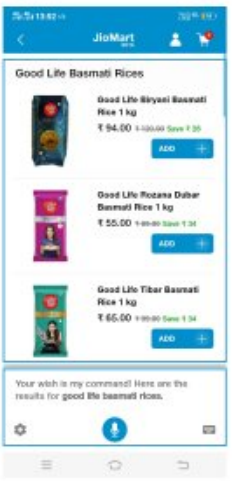
Input via Keyboard









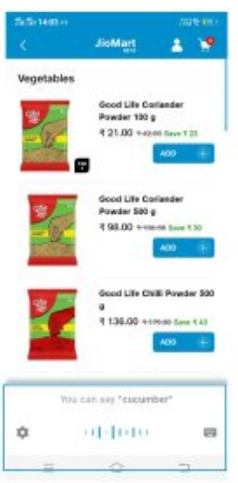


Through voice search assistant not be able to add products in the cart through voice command

To search fresh vegetable







When **user's input**- onion

Voice output- Sorry i could not find onion in stock

guided approach: What else you want to buy
you can say potato

When use provide input potato then potato is also not in stock

Identified Issues

- When user try to buy fresh vegitable then tell us vegitables are not available at this location
- Should have proper categorisation for relative products

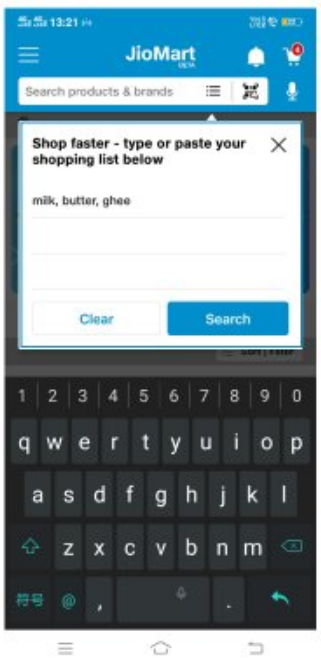
Home Page : List Feature

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
List feature	#2: Match between system and the real world #3: User control and freedom	S3	E2

User Goal:

To search product from list feature

Elaboration of Problem:



In JioMart there is "Shop Faster" options which allows user to search all the items required by them in one go.



The List option does not serve its purpose and only shows result of the first element entered in the list which can easily be performed by the Search bar therefore its functionality is redundant.

Once users have entered a list, they cannot add or remove items from them. They can only start a new search, or search individually, but cannot rewrite on the ongoing list.

Identified Issues

1.The results are not shown properly, Only the first product is shown and highlighted

2.User should have the freedom to add and remove products from the list

Top Deals

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Top Deals	#5 : Error prevention	S2	E1

User Goal:

To add product from top deals

Elaboration of Problem:



There is no notification when a user is not able to add an item to a desired quantity.

Only the "+" button stops responding without giving an indication as to why another quantity cannot be added.

Identified Issues

Top Deals on Home page should notify the user when a desired quantity of a product cannot be further added to the cart.

Recommendations:

There should be a visual cue or disclaimer
The quantity, price and discount should be clearly highlighted.
The visual hierarchy of the product name, discount, price and quantity should be relooked
Quantity should be separately added and clearly visible for user to make decisions

Address

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Address	#6 :Recognition rather than recall	S2	E1

User Goal:

To search for an entity.

Elaboration of Problem:



In JioMart only pin code is mentioned in the delivery location detail.

Recognition rather than recall

Using the name/location of the user to Personalize the delivery location details.



In JioMart mobile application automatically detecting the location is not enabled whereas in the jiomart.com website has this feature.

Identified Issues

Automatic Detection of Location via GPS should be enabled.

The user manually types the location pin code and there is no feature of automatic location update via GPS.
(for example, adding the feature "use my current location" for capturing the location via GPS can be added.)

Customer support

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Help & Support	#4: Consistency and standards #10:Help and documentation	S4	E3

User Goal:

Want to resolve Payment issues
Want to know order delivery date
Raise complaint
Contact customer care
Deals and offers

Elaboration of Problem:







Identified Issues

Increase mental load to find information from FAQs

Difficult to find complaint section because of placement of complaint section

Recommendations:

Design of the FAQs section should be relook
Complaint section should be upfront to the users so that ease to able to find for complaint

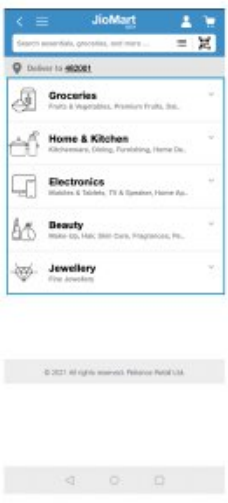
Shop by category (hamburger menu + all categories top navigational option)

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Menu & Categories	#2: Match between system and real world #4: Consistency and standards #6: Recognition rather than recall #8: Aesthetic and minimalist design	S2	E1

User Goal:

To browse product by select category and subcategory

Elaboration of Problem:





Identified Issues

Hierarchy of the product category is not aligned with main category

Some product category have not placed within the main category (Missing of product category) like fashion, medicine, etc

Main category have used icons and sub category used images instead of icon that break visual consistency

Old icons used for the main category of the product thats why difficult to recognise product category through the icons

Principle(s) Violated:

Match between system and real world
Consistency and standards
Recognition rather than recall
Aesthetic and minimalist design

Recommendations:

Hierarchy of product category should be aligned at every place on JioMart
Product category list should be equal
Should be used image instead of icon for product category and product for better recognisation of product category

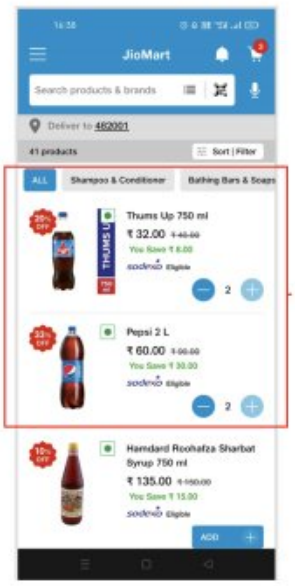
Product Listing page

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
PLP	#5 :Error prevention	S2	E2

User Goal:

To browse and add products to cart

Elaboration of Problem:



When a user cannot add more than a certain quantity then only the "+" button stops responding.

Suggestion:

A notification specifying the user an appropriate message such as "maximum quantity reached" etc. can be added.

Filter and sort

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
Filter & sort	#10:Help users recognize, diagnose, and recover from errors	S2	E2

User Goal:

To filter and sort on product

Elaboration of Problem:





The red dot creates ambiguity





User uses Search to find something specific from large amounts of information. They usually use keywords or type in phrases to find what they're looking for. It is the most common way of finding things on a app.

Filter is used when they want to narrow the list of available items based on specific criteria such as brand, price range or colour.

Sort is used when they don't want to narrow their search but rather, arrange it in a certain order, for example, newest first or alphabeticall

Problem identified

Selected filter has low visibility and has poor visual identification.

Product description page

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
PDP	#5: Error prevention #3: User control and freedom #4: Consistency & standard	S3	E2

User Goal:

To browse offers ,info of the products and add product to cart

Elaboration of Problem:



End of scroll

Problem Area	Heuristic(s) Violated	Severity Rating	Ease of Fixing
End of scroll	#7:Flexibility and efficiency of use	S1	E1

User Goal:

To browse offers and products

Elaboration of Problem:



Abrupt ending user has not clue of what to do next?